Keshav Sharma

India | 8168289407 | keshavsharma71982@gmail.com | LinkedIn Link | GitHub Link | Portfolio

EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, SQL, Java, C/C++
Quantitative & Statistical Analysis: Statistical Analysis, Probability, Data Analysis, Exploratory Data Analysis, Hypothesis Testing, Time-Series Analysis, Risk Analysis
Financial & Market Analytics: Financial Data Analysis, Market Data Analysis, Technical Analysis, Anomaly Detection, Quantitative Modelling
Machine Learning: Scikit-learn, Pandas, NumPy, Feature Engineering, Ensemble Methods, Model Evaluation
Deep Learning: PyTorch, TensorFlow/Keras, LSTM, Neural Networks
Algorithms & Data Structures: Data Structures & Algorithms, Algorithm Design, Optimisation, Complexity Analysis
Data Visualisation: Power BI, Matplotlib, Plotly
Databases & Tools: SQL, SQLite, MySQL, MongoDB, Redis, Git/GitHub, Jupyter Notebook

EXPERIENCE

VOIS – Vodafone Idea Foundation* |LINK** — ***Aug – Sep, 2025

- Analysed 2 real-world business datasets containing 8,500+ records, applying data preprocessing and AI-assisted analytics to identify trends and generate 15 actionable business insights for data-driven decision-making.
- Validated analytical findings for accuracy and consistency, reviewing outputs to ensure reliable and meaningful insights for business recommendations.
- Established a standardised analytics workflow covering data preprocessing, insight generation, and validation, enabling consistent conversion of raw business data into actionable recommendations.

PROJECTS

***Real-Time Stock Market Anomaly Detector* |LINK**

- Engineered a real-time market analytics pipeline processing 2M+ NSE records, combining price signals and financial-news sentiment to detect unusual market behaviour.
- Developed and evaluated Isolation Forest and LSTM Autoencoder models, achieving 88% precision and 85% recall on validation data for anomaly detection.
- Optimised the analytics pipeline to reduce detection latency by 35%, enabling near real-time monitoring and faster identification of unusual market events.

***Financial Transaction Anomaly Detection & Audit Dashboard* | LINK**

- Engineered an end-to-end financial analytics pipeline processing 50K+ PaySim transactions, combining statistical outlier analysis, duplicate detection, vendor-velocity checks, HHI concentration analysis, and Isolation Forest for transaction-risk detection.
- Designed a star-schema SQLite data warehouse and composite 0–100 risk scoring framework, identifying 357 high/critical-risk transactions with end-to-end analysis completed in under 7 seconds.
- Built a 5-page Power BI audit dashboard with 20+ DAX measures to surface vendor risk, anomaly rates, transaction trends, and risk insights, supported by automated pipeline tests.

***Customer Lifetime Value Engine* |LINK**

- Built a probabilistic model (BG/NBD + Gamma-Gamma) on 1M+ historical transaction records to forecast customer lifetime value directly analogous to modelling player value and monetisation potential.
- Segmented customers into value-based cohorts to identify which segments drive the most revenue, supporting prioritisation.
- Identified the top 20% of customers driving ~65% of forecasted revenue, visualised through Power BI dashboards for stakeholder-facing reporting.

***RedisLite* |LINK**

- Engineered a Redis-inspired in-memory data store in Python with a custom RESP protocol parser, multithreaded TCP server, and an O(1) LRU cache with lazy/active TTL expiration — core data pipeline infrastructure for high-throughput storage and retrieval.
- Designed a four-layer architecture (transport, storage engine, cache/TTL, persistence) implementing Write-Ahead Logging (WAL) and automatic crash recovery to ensure data integrity.
- Validated the system with 46 automated PyTest unit tests, benchmarking 619K SET ops/sec in-memory and 2.8K ops/sec with durable WAL persistence.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- Deloitte Australia Data Analytics Job Simulation |**LINK** 2025